

SE - Database Management – SQL & PL/SQL

Consider a scenario in which you have to create a table called Nobel_win where you have given the information about the entity who has achieved nobel prize in a particular subject all over the world.

Over here the table Nobel_win will contain attributes like year, subject, winner, country and category

You need to use the sql commands to create the table given and to solve the problem given below.

Eg :

YEAR	SUBJECT	WINNER	COUNTRY	CATEGORY
1970	Physics	Hannes Alfven	Sweden	Scientist
1970	Physics	Louis Neel	France	Scientist
1970	Chemistry	Luis Federico Leloir	France	Scientist
1970	Physiology	Ulf von Euler	Sweden	Scientist
1970	Physiology	Bernard Katz	Germany	Scientist
1970	Literature	Aleksandr Solzhenitsyn	Russia	Linguist
1970	Economics	Paul Samuelson	USA	Economist
1970	Physiology	Julius Axelrod	USA	Scientist
1971	Physics	Dennis Gabor	Hungary	Scientist
1971	Chemistry	Gerhard Herzberg	Germany	Scientist
1971	Peace	Willy Brandt	Germany	Chancellor
1971	Literature	Pablo Neruda	Chile	Linguist
1971	Economics	Simon Kuznets	Russia	Economist
1978	Peace	Anwar al-Sadat	Egypt	President
1978	Peace	Menachem Begin	Israel	Prime Minister
1987	Chemistry	Donald J. Cram	USA	Scientist
1987	Chemistry	Jean-Marie Lehn	France	Scientist
1987	Physiology	Susumu Tonegawa	Japan	Scientist
1994	Economics	Reinhard Selten	Germany	Economist
1994	Peace	Yitzhak Rabin	Israel	Prime Minister
1987	Physics	Johannes Georg Bednorz	Germany	Scientist
1987	Literature	Joseph Brodsky	Russia	Linguist
1987	Economics	Robert Solow	USA	Economist
1994	Literature	Kenzaburo Oe	Japan	Linguist

Now make sure to solve the following query :

Ans :

Create Table :

```
CREATE TABLE Nobel_win (  
    year INT, subject VARCHAR(50),  
    winner VARCHAR(100),  
    country VARCHAR(50),  
    category VARCHAR(50)  
);
```

Insert Sample Data :

```
INSERT INTO Nobel_win VALUES (1970,'Physics','Hannes  
Alfven','Sweden','Scientist');
```

```
INSERT INTO Nobel_win VALUES (1970,'Physics','Louis  
Neel','France','Scientist');
```

```
INSERT INTO Nobel_win VALUES (1970,'Chemistry','Luis Federico  
Leloir','France','Scientist');
```

```
INSERT INTO Nobel_win VALUES (1970,'Physiology','Ulf von  
Euler','Sweden','Scientist');
```

```
INSERT INTO Nobel_win VALUES (1970,'Physiology','Bernard  
Katz','Germany','Scientist');
```

```
INSERT INTO Nobel_win VALUES (1970,'Literature','Aleksandr  
Solzhenitsyn','Russia','Linguist');
```

```
INSERT INTO Nobel_win VALUES (1970,'Economics','Paul  
Samuelson','USA','Economist');
```

```
INSERT INTO Nobel_win VALUES (1970,'Physiology','Julius  
Axelrod','USA','Scientist');
```

```
INSERT INTO Nobel_win VALUES (1971,'Physics','Dennis  
Gabor','Hungary','Scientist');
```

```
INSERT INTO Nobel_win VALUES (1971,'Chemistry','Gerhard  
Herzberg','Germany','Scientist');
```

```
INSERT INTO Nobel_win VALUES (1971,'Peace','Willy  
Brandt','Germany','Chancellor');
```

```
INSERT INTO Nobel_win VALUES (1971,'Literature','Pablo  
Neruda','Chile','Linguist');
```

```
INSERT INTO Nobel_win VALUES (1971,'Economics','Simon  
Kuznets','Russia','Economist');
```

```
INSERT INTO Nobel_win VALUES (1978,'Peace','Anwar al-  
Sadat','Egypt','President');
```

INSERT INTO Nobel_win VALUES (1978,'Peace','Menachem Begin','Israel','Prime Minister');

INSERT INTO Nobel_win VALUES (1987,'Chemistry','Donald J. Cram','USA','Scientist');

INSERT INTO Nobel_win VALUES (1987,'Chemistry','Jean-Marie Lehn','France','Scientist');

INSERT INTO Nobel_win VALUES (1987,'Physiology','Susumu Tonegawa','Japan','Scientist');

INSERT INTO Nobel_win VALUES (1987,'Physics','Johannes Georg Bednorz','Germany','Scientist');

INSERT INTO Nobel_win VALUES (1987,'Literature','Joseph Brodsky','Russia','Linguist');

INSERT INTO Nobel_win VALUES (1987,'Economics','Robert Solow','USA','Economist');

INSERT INTO Nobel_win VALUES (1994,'Economics','Reinhard Selten','Germany','Economist');

INSERT INTO Nobel_win VALUES (1994,'Peace','Yitzhak Rabin','Israel','Prime Minister');

INSERT INTO Nobel_win VALUES (1994,'Literature','Kenzaburo Oe','Japan','Linguist');

year	subject	winner	country	category
1970	Physics	Hannes Alfven	Sweden	Scientist
1970	Physics	Louis Neel	France	Scientist
1970	Chemistry	Luis Federico Leloir	France	Scientist
1970	Physiology	Ulf von Euler	Sweden	Scientist
1970	Physiology	Bernard Katz	Germany	Scientist

year	subject	winner	country	category
1970	Literature	Aleksandr Solzhenitsyn	Russia	Linguist
1970	Economics	Paul Samuelson	USA	Economist
1970	Physiology	Julius Axelrod	USA	Scientist
1971	Physics	Dennis Gabor	Hungary	Scientist
1971	Chemistry	Gerhard Herzberg	Germany	Scientist
1971	Peace	Willy Brandt	Germany	Chancellor
1971	Literature	Pablo Neruda	Chile	Linguist
1971	Economics	Simon Kuznets	Russia	Economist
1978	Peace	Anwar al-Sadat	Egypt	President
1978	Peace	Menachem Begin	Israel	Prime Minister
1987	Chemistry	Donald J. Cram	USA	Scientist
1987	Chemistry	Jean-Marie Lehn	France	Scientist
1987	Physiology	Susumu Tonegawa	Japan	Scientist
1987	Physics	Johannes G. Bednorz	Germany	Scientist
1987	Literature	Joseph Brodsky	Russia	Linguist
1987	Economics	Robert Solow	USA	Economist
1994	Economics	Reinhard Selten	Germany	Economist
1994	Peace	Yitzhak Rabin	Israel	Prime Minister
1994	Literature	Kenzaburo Oe	Japan	Linguist

● Write sql query to find the nobel prize winners of the year 1970.
Return year,subject and winner.

```
SELECT year, subject, winner
FROM Nobel_win
WHERE year = 1970;
```

OutPut :

year	subject	winner
1970	Physics	Hannes Alfven
1970	Physics	Louis Neel
1970	Chemistry	Luis Federico Leloir
1970	Physiology	Ulf von Euler
1970	Physiology	Bernard Katz
1970	Literature	Aleksandr Solzhenitsyn
1970	Economics	Paul Samuelson
1970	Physiology	Julius Axelrod

● Write sql query to find the nobel prize winners in chemistry between the years 1965 and 1975. Begin and end values are includedReturn year, subject, winner and country.

```
SELECT year, subject, winner, country
FROM Nobel_win
```

WHERE subject = 'Chemistry'

AND year BETWEEN 1965 AND 1975;

OutPut :

year	subject	winner	country
1970	Chemistry	Luis Federico Leloir	France
1971	Chemistry	Gerhard Herzberg	Germany

● Write sql query to retrieve the details of the winners whose first name matches with the string 'Louis'. Return year,subject,winner,country

SELECT year, subject, winner, country

FROM Nobel_win

WHERE winner LIKE 'Louis';

WHERE LOWER(winner) LIKE 'louis';

OutPut :

year	subject	winner	country
1970	Physics	Louis Neel	France

● Write sql query to find Nobel prize winners for the subject that does not begin with the letter 'P'. Order the result by year, descending and winner in ascending

SELECT year, subject, winner, country

FROM Nobel_win

WHERE subject NOT LIKE 'P'

ORDER BY year DESC, winner ASC;

OutPut :

year	subject	winner	country
1994	Economics	Reinhard Selten	Germany
1994	Literature	Kenzaburo Oe	Japan
1994	Peace	Yitzhak Rabin	Israel
1987	Chemistry	Donald J. Cram	USA
1987	Chemistry	Jean-Marie Lehn	France
1987	Economics	Robert Solow	USA
1987	Literature	Joseph Brodsky	Russia
1978	Peace	Anwar al-Sadat	Egypt
1978	Peace	Menachem Begin	Israel
1971	Chemistry	Gerhard Herzberg	Germany
1971	Economics	Simon Kuznets	Russia
1971	Literature	Pablo Neruda	Chile
1971	Peace	Willy Brandt	Germany
1970	Chemistry	Luis Federico Leloir	France

year	subject	winner	country
1970	Economics	Paul Samuelson	USA
1970	Literature	Aleksandr Solzhenitsyn	Russia

● Write sql query to find the details of 1970 Nobel prize winners. Order the result by subject, ascending except for 'Chemistry' and 'Economics' which will come at the end of the result set. Return year, subject, winner , country and category

```

SELECT year, subject, winner, country, category
FROM Nobel_win
WHERE year = 1970
ORDER BY
CASE
    WHEN subject IN ('Chemistry', 'Economics') THEN 1
    ELSE 0
END,
subject ASC;

```

OutPut:

year	subject	winner	country	category
1970	Literature	Aleksandr Solzhenitsyn	Russia	Linguist

year	subject	winner	country	category
1970	Physiology	Ulf von Euler	Sweden	Scientist
1970	Physiology	Bernard Katz	Germany	Scientist
1970	Physiology	Julius Axelrod	USA	Scientist
1970	Physics	Hannes Alfven	Sweden	Scientist
1970	Physics	Louis Neel	France	Scientist
1970	Chemistry	Luis Federico Leloir	France	Scientist
1970	Economics	Paul Samuelson	USA	Economist